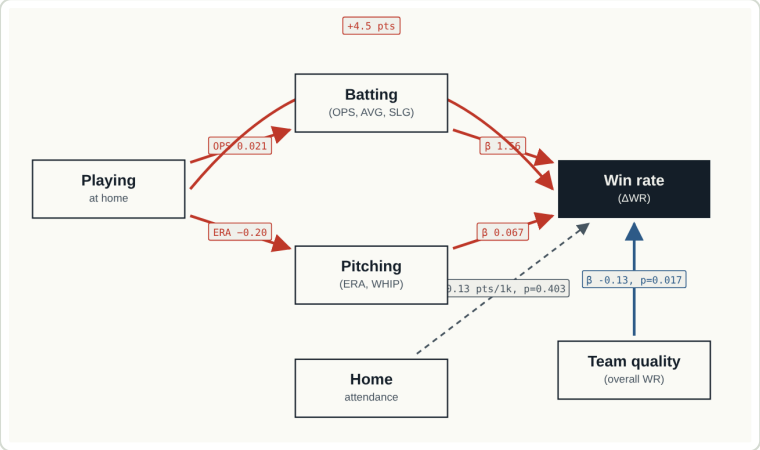
Three System Maps Comparison Report

Each map answers the same question: **why do KBO home teams win about 4.5 points more often?** Map A uses only our original hypotheses, Map B adds theory and hidden non-baseball features, and Map C removes 15 confounders with machine learning. **Solid arrows** are supported by a test at α = 0.05; **dashed arrows** were not significant or could not be measured. Data: 152 team-seasons (2010–2025) and 3,494 games (2021–2025).

MAP A · STATISTICAL SIGNIFICANCE (ORIGINAL HYPOTHESES)

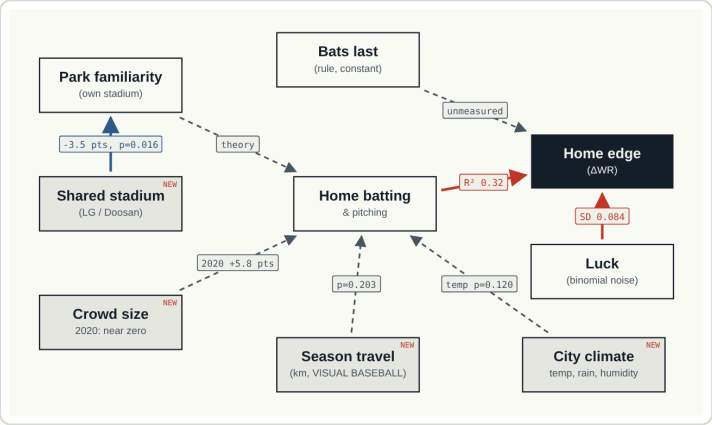


ARROW	EVIDENCE	STATUS
Home → batting	OPS +0.022, p<0.001	**Solid**
Home → pitching	ERA −0.20, p<0.001	**Solid**
Batting + pitching → ΔWR	R^2 = 0.32	**Solid**
Home → ΔWR (total)	+4.5 pts, p<0.001	**Solid**
Attendance → ΔWR	−0.13/1k, p=0.40	**Dashed (n.s.)**
Team quality → ΔWR	−0.13, p=0.017*	**Solid (−)**

* Simple regression. With attendance and season added, p = 0.058, so we read it as a ceiling tendency: a team that wins everywhere has less room to win more at home.

Reading: the edge is real and works through on-field performance. We found no significant support for the crowd half of H_1; that is absence of evidence, not proof of no effect.

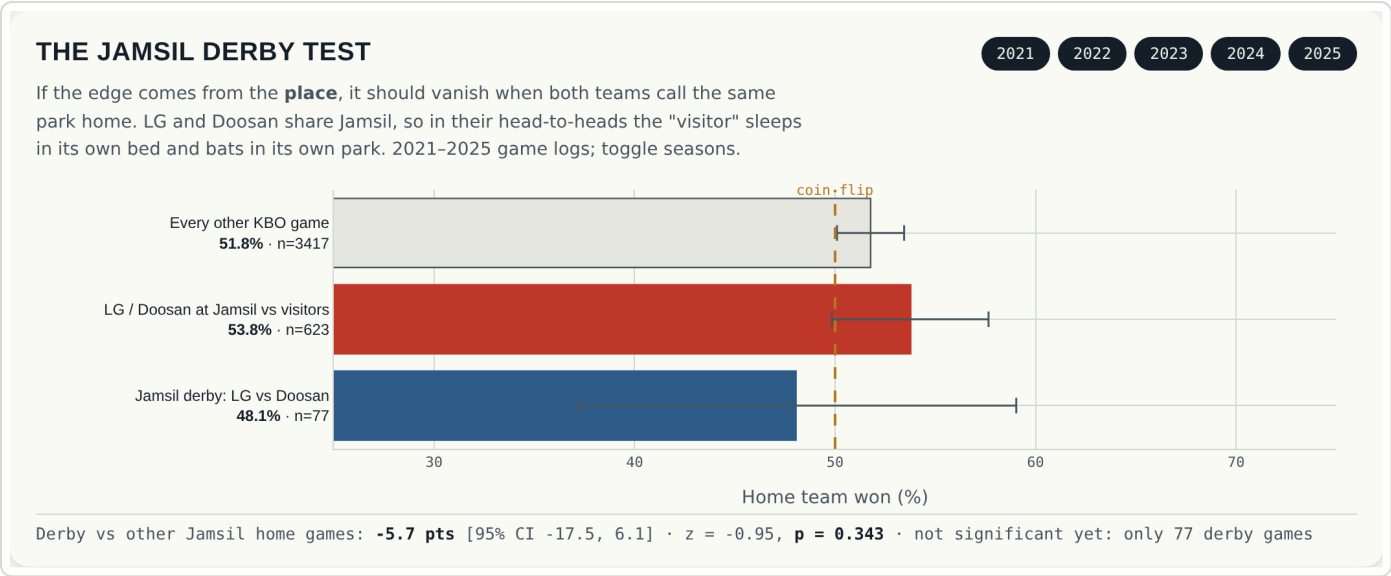
MAP B · THEORY, SCENARIOS AND HIDDEN FEATURES



ARROW	EVIDENCE	STATUS
Shared stadium → familiarity	−3.5 pts, p=0.016 (Holm 0.20)	**Solid, nominal only**
Crowd → performance	2020 edge +5.8, intact	**Weakened**
Season travel	p=0.20	**Dashed**
City climate	temp p=0.12	**Dashed**
Bats last (rule)	constant	**Unmeasured**
Luck → ΔWR	SD 8.4 vs 7.4 obs.	**Solid**

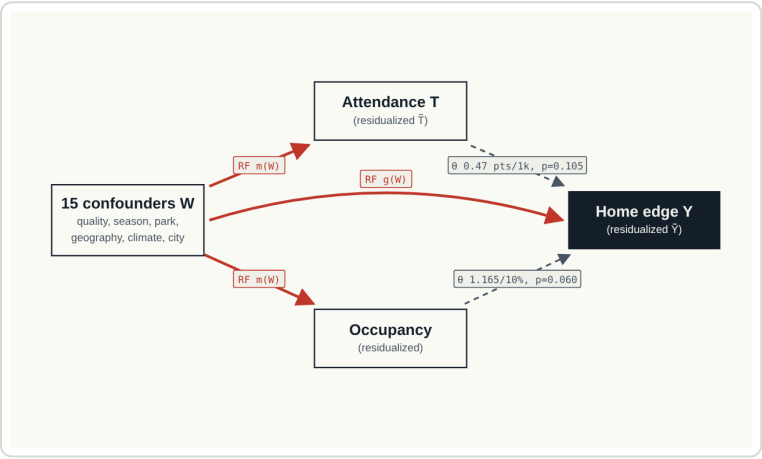
The shared-stadium arrow is significant on its own but not after Holm correction for the 22 tests we ran, so we treat it as exploratory.

Scenario tested: “What if the edge is about the place, not the people?” Two natural experiments support it: near-empty stands in 2020 (closed until late July, then partly open) left the edge intact, and in the **Jamsil derby** (LG vs Doosan, both at home) the home side won only 48.1% of 77 games vs 53.8% when the same clubs host visitors at Jamsil (p = 0.34).



Jamsil derby test from the webpage (2021–2025 game logs). Direction supports the place explanation; the sample is too small for significance.

MAP C · MACHINE LEARNING (DOUBLE ML)



Random forests predict attendance and ΔWR from 15 confounders (team quality, season, stadium, geography, city, climate). The leftovers are regressed on each other (partialling-out, 5-fold cross-fitting × 20 repeats).

EFFECT AFTER REMOVING 15 CONFOUNDERS	θ [95% CI]	P
Attendance (per 1,000 fans)	+0.47 [−0.10, +1.03]	0.105
Occupancy (per +10%)	+1.2 [−0.05, +2.4]	0.060
Random forest, all 17 features	cross-validated R^2 = −0.11	—

Reading: no reliable crowd effect even with flexible controls, and the home edge cannot be predicted from team-level features at all, consistent with Map B's finding that most team-to-team variation is luck.

COMPARING THE THREE MAPS

	MAP A · STATISTICAL	MAP B · THEORY & HIDDEN FEATURES	MAP C · MACHINE LEARNING
Question	Do our original variables matter?	What else could cause the edge?	Is the crowd effect hidden by confounders?
Method	Paired / one-sample t, simple OLS	Welch t, natural experiments, χ^2 luck test; Holm correction across all 22 tests	Double ML, random forest CV
Adds	Mechanism: batting and pitching	Place over people; luck as a cause of variation	Robustness to 15 confounders at once
Crowd verdict	**Not significant (p = 0.40)**	**Weakened by 2020; shared-stadium and derby results point the same way but neither survives correction**	**Not significant (p = 0.10)**
Main limit	One X at a time; omitted variables	Team-level features vary across only 10 clubs	Black-box nuisance models; n = 152

Convergence. Three methods with different assumptions point the same way: the home advantage is real and runs through on-field performance, and none of them finds a significant crowd effect. The ballpark explanation fits the evidence best but remains a hypothesis, because its strongest direct test (the Jamsil derby) is underpowered. Agreement across methods is what lets us move from statistical results (Levels 4–5) to a reasoned explanation (Level 6).

JUSTIFICATION DOCUMENT

A. Why some “non-significant” variables might still matter

VARIABLE	OUR RESULT	WHY IT MAY STILL MATTER	WHAT WOULD SETTLE IT
Attendance	`−0.13 pts/1k, CI [−0.42, +0.17]`	Not significant is not the same as zero. The CI still allows up to +0.17 pts per 1,000 fans, i.e. about +2.6 pts between a 5,000 and a 20,000 crowd. Crowd effects may also be **nonlinear** (only at sell-outs) or **offsetting**: Jeong (2023) found that crowd limits raised home batting but hurt home pitching, which cancel out in win rate.	Game-level models with sell-out thresholds; separate batting and pitching outcomes.
Occupancy	`DML +1.2 pts per +10%, p = 0.06`	Close to significance and its sign **flips** from negative (simple OLS) to positive once confounders are removed. That pattern suggests suppression: the two biggest-crowd clubs share a park and have a smaller edge, which hides any crowd benefit in naive models.	More seasons; occupancy measured against real seat counts per season.
Jamsil derby	`48.1% vs 53.8%, p = 0.34`	The effect is in the predicted direction and practically large (−5.7 pts), but 77 games give very low power. Detecting this gap at 80% power needs about **1,200 derby games**, roughly 75 seasons at 16 per year.	Pool more seasons; add a third shared park if one appears.
Season travel	`simple p = 0.20; multi p = 0.049`	Significance depends on which controls are included, so the effect is fragile rather than absent. Korea's short distances keep it small on average, but long road trips on short rest could still matter for specific series.	Game-level rest days and consecutive road games.
Weather and calendar	`0 of 12 features, p > 0.17`	Averages can hide interactions, e.g. heat may affect visiting pitchers more in open-air parks than in the Gocheok dome. A null on the average is not a null in every subgroup.	Interaction models (weather × dome, weather × starter).

B. Policy implications if we ignore these hidden features

WHO	IF HIDDEN FEATURES ARE IGNORED…	BETTER DECISION WITH THEM
Club front offices	Without the shared-stadium variable, the two biggest-crowd clubs (LG, Doosan) pull the crowd slope negative, and an analyst could conclude that fans hurt performance, or that marketing spend buys wins.	Fund attendance marketing for revenue and fan experience; do not expect it to buy wins.
KBO and the city of Seoul	Stadium plans for Jamsil (redevelopment or temporary relocation) would be evaluated only on capacity and revenue, ignoring that sharing a park removes about 3.5 pts of home edge from two clubs.	Include competitive balance in stadium decisions and track both clubs' home edge before and after any move.
Forecasters, media, bettors	Models that add crowd size or a team's past home record treat noise as signal. Team “fortress” labels are mostly luck (χ^2 p = 0.99).	Use a near-constant home bonus of about 4–5 pts; drop crowd size as a predictor.
Emergency policy (e.g. pandemics)	Leagues may fear that empty-stadium games are unfair to home teams and delay restarts or redesign schedules.	2020 evidence shows competitive fairness survived near-empty stands; decisions can focus on health and revenue.
Multiple testing	Running 22 tests and reporting only the significant ones would overstate the evidence: the shared-stadium result (p = 0.016) loses significance after Holm correction.	Report corrected p-values next to raw ones (done on our webpage) and pre-register the main test.
Researchers	Treating “not significant” as “no effect” closes questions that are only underpowered (derby, occupancy).	Report CIs and power alongside p-values; pool more seasons before concluding.

Provenance: game rows (2021–2025) carry game ID, date, teams and score and are rebuilt by `analysis/00_build_games.py` from kbo-crowd (commit 1d28725); weather is measured KMA ASOS hourly data at the first-pitch hour; occupancy is estimated. Sources: KBO home/away records and splits (Yagoonara, MyKBO Stats); KBO attendance; kbo-crowd game logs with KMA ASOS weather (2021–2025); Wikipedia/KMA stadium, census and climate data; Jeong, G. (2023), *Effects of no-crowd and partial-crowd games on home advantage in the KBO*, Sungkyunkwan University master's thesis. All numbers reproduce from the code repository.